

# CRASH COURSE

— GATE 2026 —

COMPUTER SCIENCE & IT

Computer Networks 04

Transport Layer  
[TCP Connections]

By - Abhishek Sir





# Recap of previous lecture

- 1 Transport Layer Services
- 2 UDP
- 3 TCP

# Topics to be covered

- 1 TCP Connection Establishment  $\Rightarrow$  V.I.M.P.  
24, 25, 26?
- 2 TCP Data Transfer



# ABOUT ME

Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : [https://t.me/abhisheksirCS\\_PW](https://t.me/abhisheksirCS_PW)





## Topic : Reference



→ Request for Comments (RFC)

by Internet Engineering Task Force (IETF)



## Topic : Syllabus

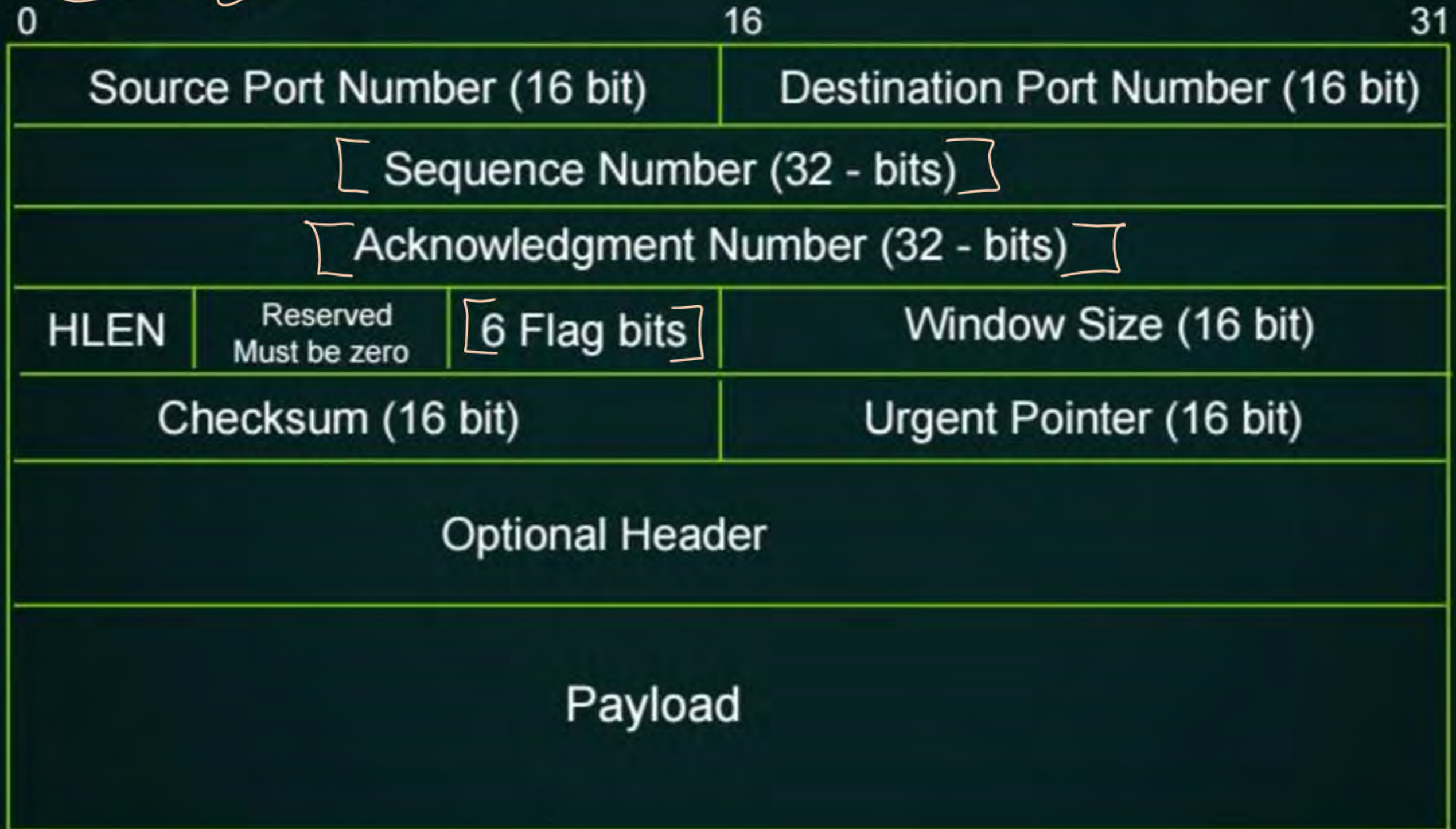


↓  
**Transport layer** : flow control and congestion control, UDP, TCP, sockets;





## Topic : TCP Header





## Topic : TCP Flag Bits



- => URG : Urgent
- => ACK : Acknowledgement ✓
- => PSH : Push
- => RST : Reset
- => SYN : Synchronize ✓
- => FIN : Finish





## Topic : TCP Operation



Three phases of TCP operation :

1. Connection establishment ✓

→ 3-way handshake process that establishes a connection →  $V.V.V.$   
Imp.

2. Data transfer

3. Connection termination

→ 4-way handshake process that closes the connection →  $[CS-17]$



## Topic : TCP Connection Establishment



- Connection establishment between TCP client and TCP server
- 3-way handshake process

→ Always TCP client initiate the connection request to TCP server ← Imp.



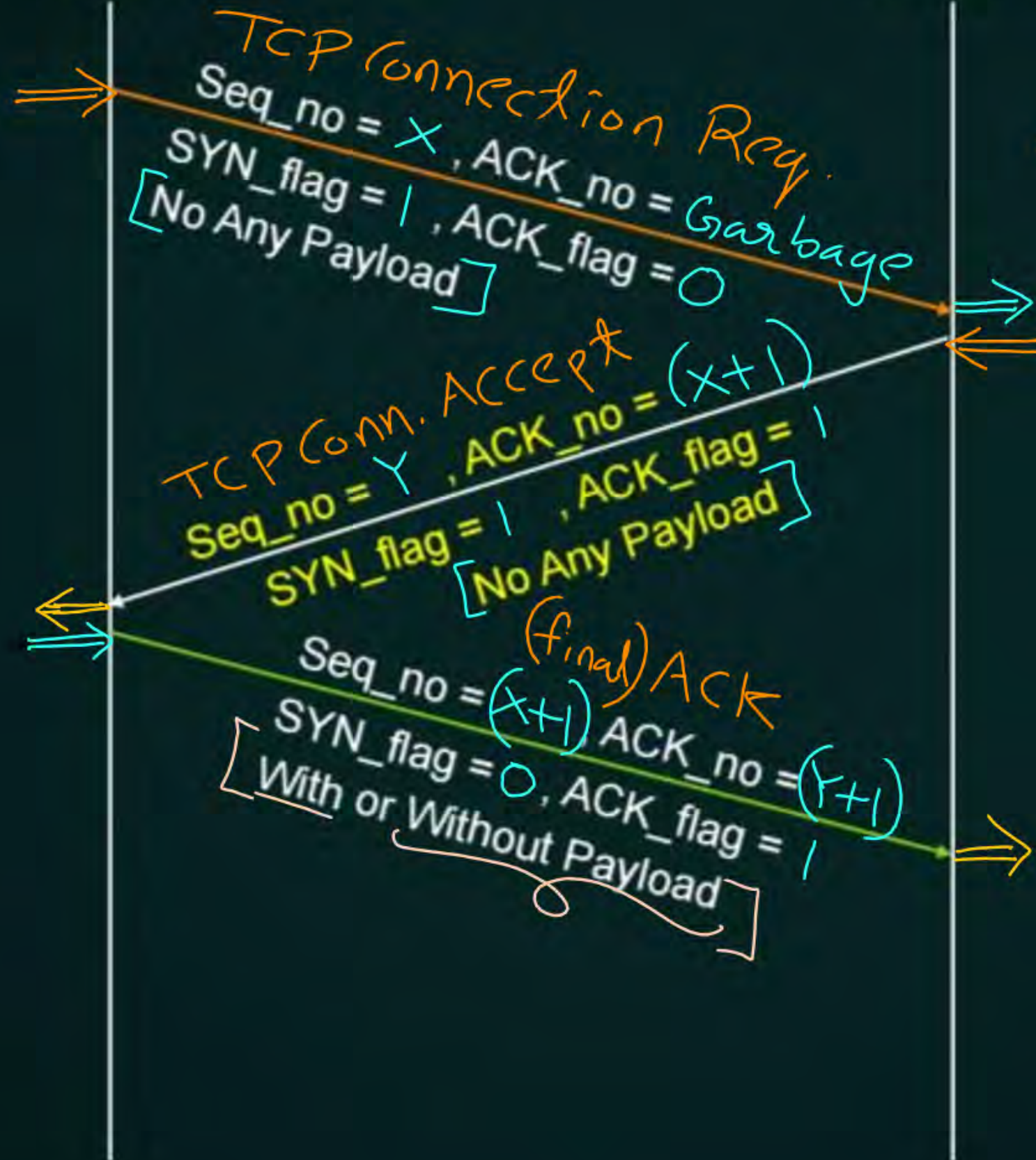
Client\_ISN =  $X$

[32 bit]  
Seq. No.]

TCP Client

TCP Server

Server\_ISN =  $Y$



\* Ready to Listen (Read)



## Topic : TCP Connection Establishment



- Initial Sequence Number : if SYN flag is on  
[both TCP client and server randomly chooses their initial sequence number,  
to prevent from some kind of attacks]





## Topic : TCP Connection Establishment



→ SYN segment do not carry user data [no any payload]

→ SYN segment consume one sequence number ⇒

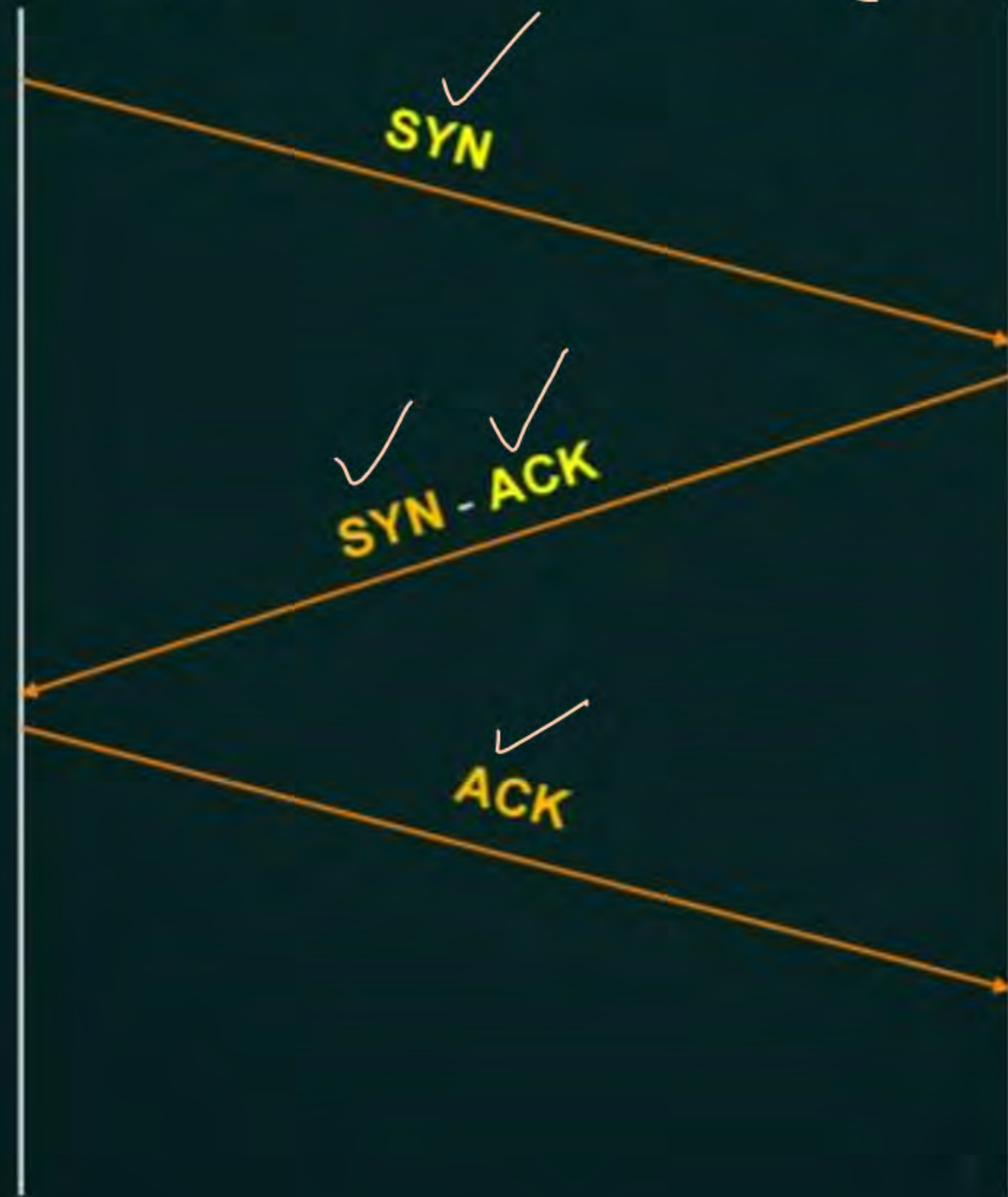


# Topic : TCP Connection Establishment



[TCP Client]

[TCP Server]





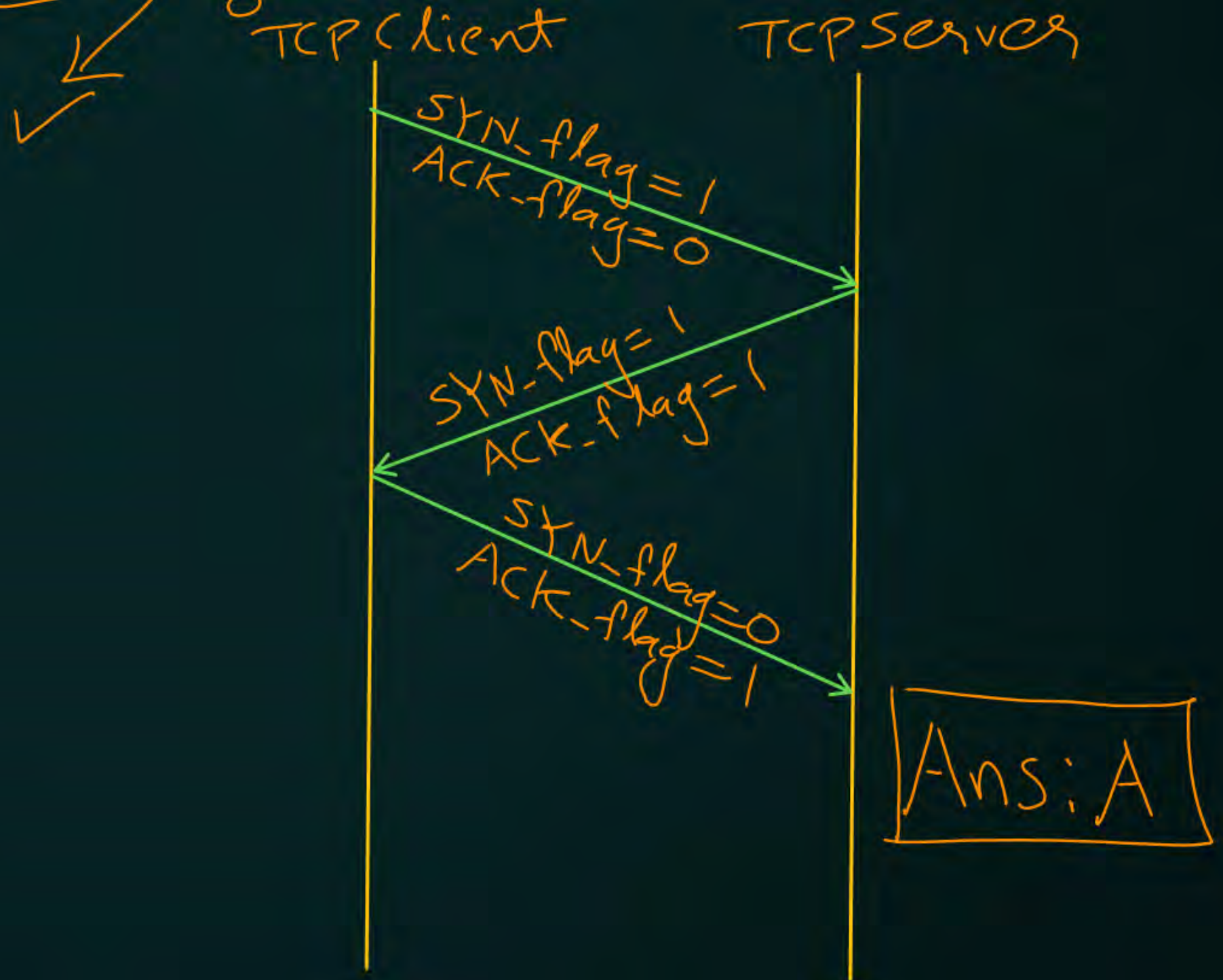
## Example 9 :-

[MSQ]



#Q. Consider the 3-way handshake process for TCP connection establishment. Which of the following option(s) is/are INCORRECT with respect to TCP header flags that are set in the packets?

- ☒ A SYN = 0, ACK = 0
- ☐ B SYN = 0, ACK = 1
- ☐ C SYN = 1, ACK = 0
- ☐ D SYN = 1, ACK = 1





**[MSQ] [GATE-2025, Set-1, 1 Mark]**

IIT-R



#Q. Consider the 3-way handshaking protocol for TCP connection establishment. Let the three packets exchanged during the connection establishment be denoted as P1, P2, and P3, in order. Which of the following option(s) is/are TRUE with respect to TCP header flags that are set in the packets?

☒ A P1: SYN = 1, ACK = 0

☒ B P2: SYN = 1, ACK = 1

☐ C P2: SYN = 0, ACK = 1

☐ D P3: SYN = 1, ACK = 1

Ans: A & B



TCP Client

TCP Server

P1 : SYN\_flag = 1 , ACK\_flag = 0

P2 : SYN\_flag = 1 , ACK\_flag = 1

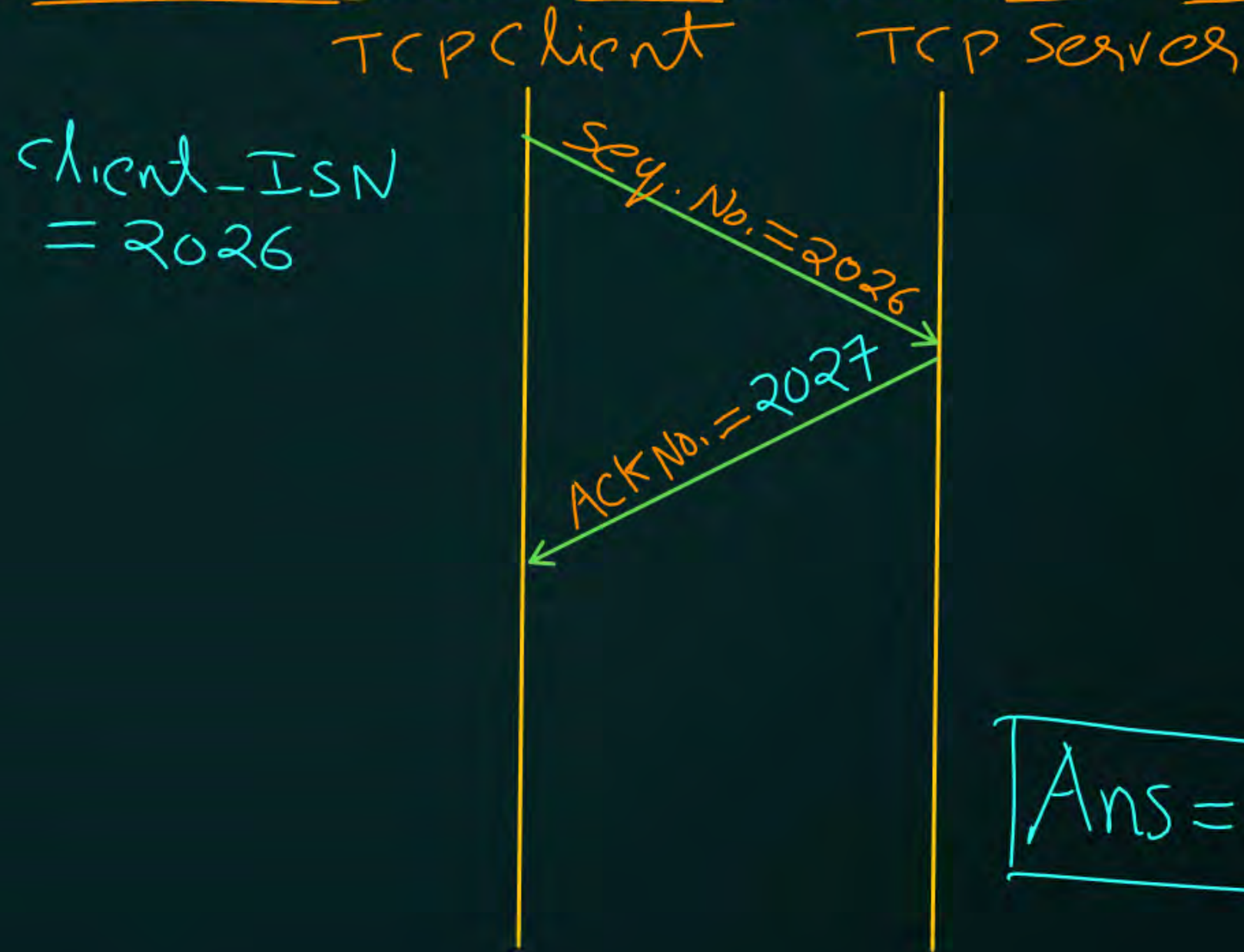
P3 : SYN\_flag = 0 , ACK\_flag = 1

## Example 10 :-

[NAT]



#Q. Consider the 3-way handshake process for TCP connection establishment. The TCP client is sending SYN segment to TCP server with sequence number 2026 (in decimal) and the TCP server is sending SYN - ACK segment to TCP client. The decimal equivalent value of acknowledgement number in the SYN - ACK segment is \_\_\_\_\_.





**[MSQ] [GATE-2024, Set-1, 1 Mark]** *11SC*

#Q. TCP client P successfully establishes a connection to TCP server Q. Let  $N_P$  denote the sequence number in the SYN sent from P to Q. Let  $N_Q$  denote the acknowledgement number in the SYN ACK from Q to P. Which of the following statements is/are CORRECT?

$$N_Q = [N_P + 1]$$

☐ A The sequence number  $N_P$  is always 0 for a new connection *False*

☒ B The sequence number  $N_P$  is chosen randomly by P *True*

☐ C The acknowledgement number  $N_Q$  is equal to  $N_P$  *False*

☒ D The acknowledgement number  $N_Q$  is equal to  $(N_P + 1)$  *True*

Ans: B & D

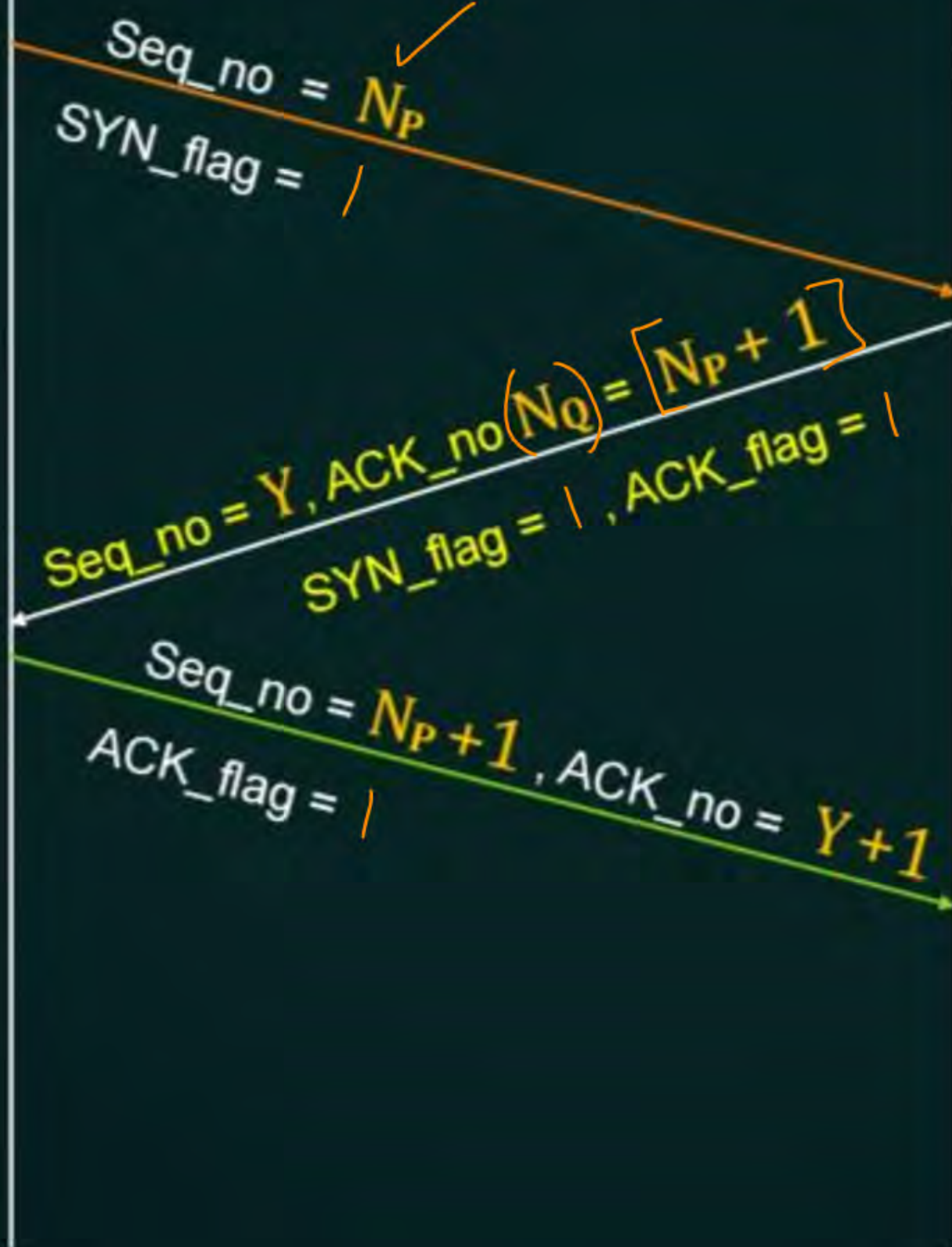


Client\_ISN =  $[N_P]$

[TCP Client] (P)

[TCP Server] (Q)

Server\_ISN =  $Y$







## Topic : TCP State Transition



→ Life-cycle of a TCP connection,

[ Changes in a TCP connection based on the exchange of data packets ]

Client\_ISN = X

TCP Client

TCP Server

Server\_ISN = Y

CLOSED

SYN, Seq\_No. = X

LISTEN

SYN\_SENT

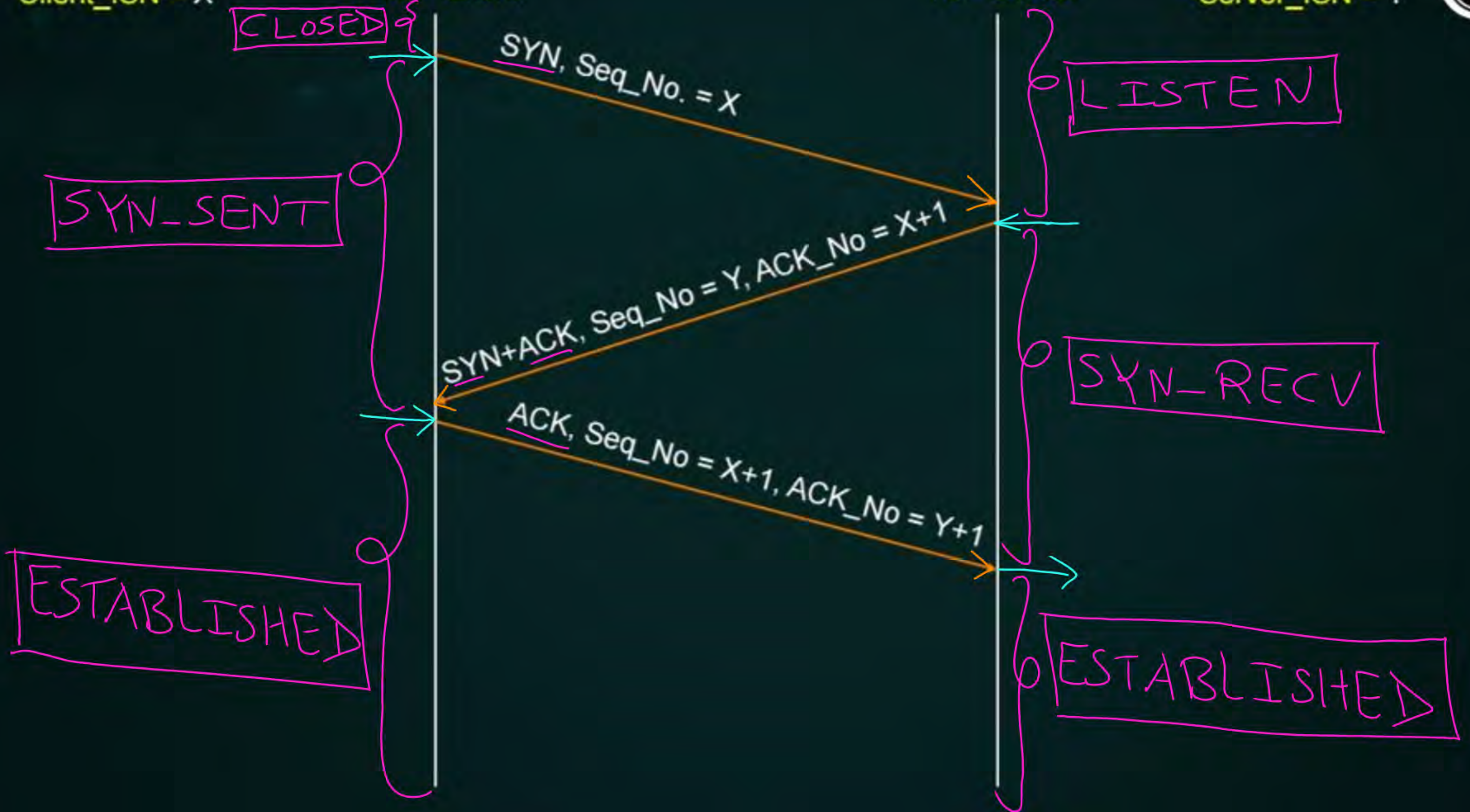
SYN+ACK, Seq\_No = Y, ACK\_No = X+1

SYN\_RECV

ACK, Seq\_No = X+1, ACK\_No = Y+1

ESTABLISHED

ESTABLISHED







## Topic : TCP

⇒ UDP Socket : sock\_DGRAM



TCP Client Process

$P_1$  ✓

SOCK\_STREAM  
(TCP Socket)

Send Buffer

Recv Buffer

TCP Server Process

$P_2$  ✓

SOCK\_STREAM  
(TCP Socket)

Send Buffer

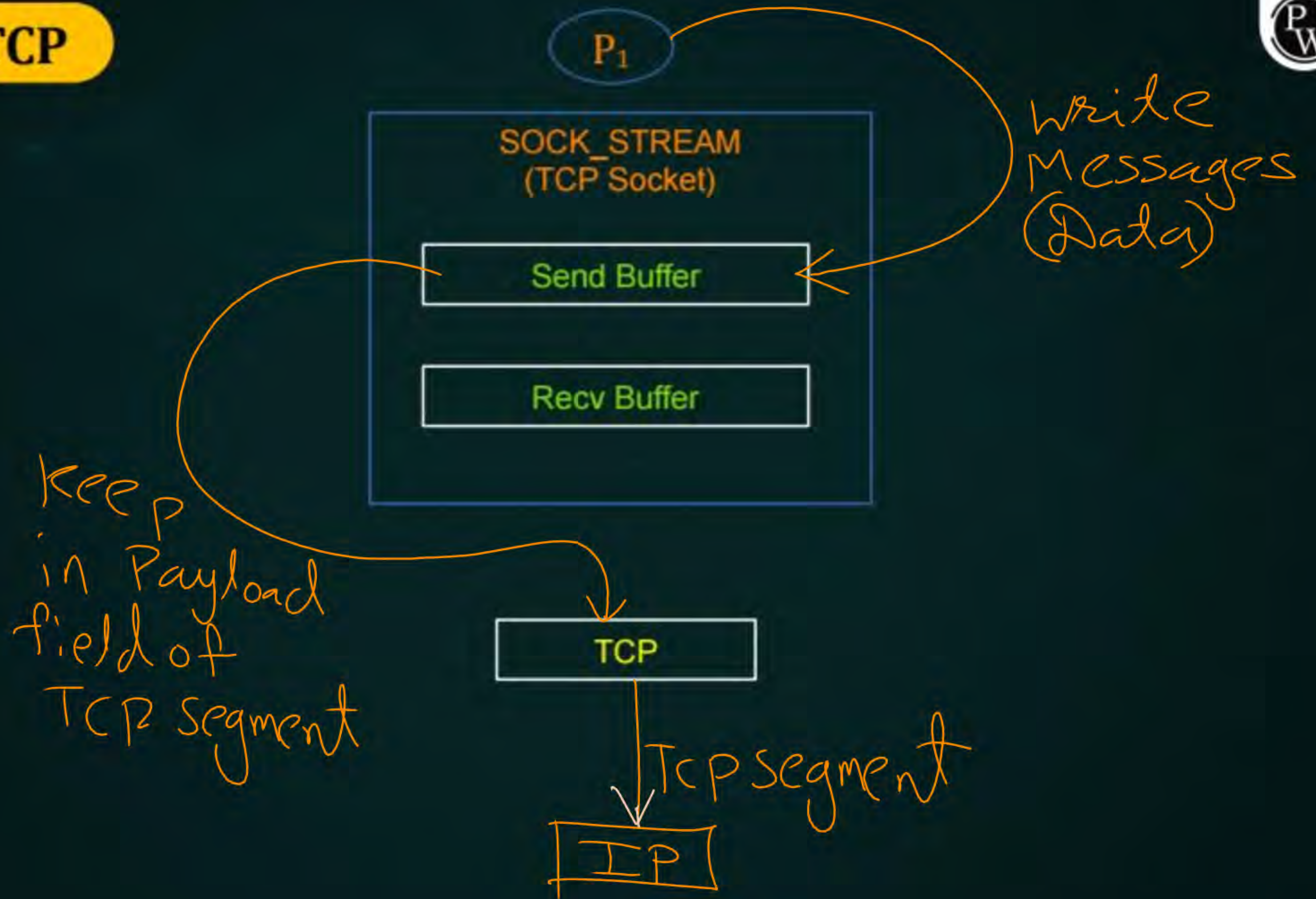
Recv Buffer

(Data or Messages)





## Topic : TCP







## Topic : TCP



Read  
Messages  
(Data)

$P_2$



TCP

IP

TCP segment

Keep  
TCP segment  
payload  
bytes

TCP Client Process

$P_1$

SOCK\_STREAM  
(TCP Socket)

LastByteWritten =  $P_1$ 's ISN

RWND =

LastByteSent =  $P_1$ 's ISN

CWND =

LastByteAcked =  $P_1$ 's ISN

Send Buffer

(MaxSendBuffer = 1000 Bytes)

Recv Buffer

(MaxRecvBuffer = 1000 Bytes)

LastByteRead =  $P_2$ 's ISN

NextByteExpected =  $P_2$ 's ISN + 1

LastByteRcvd =  $P_2$ 's ISN

⑤

③



TCP Server Process

$P_2$

SOCK\_STREAM  
(TCP Socket)

Send Buffer

(MaxSendBuffer = 1200 Bytes)

Recv Buffer

(MaxRecvBuffer = 1200 Bytes)

LastByteWritten =  $P_2$ 's ISN

LastByteSent =  $P_2$ 's ISN

LastByteAcked =  $P_2$ 's ISN

RWND =

CWND =

LastByteRead =  $P_1$ 's ISN

NextByteExpected =  $P_1$ 's ISN + 1

LastByteRcvd =  $P_1$ 's ISN





## Topic : Maximum Segment Size



→ Maximum Segment Size (MSS)

→ Maximum TCP segment payload size (in bytes)

[ Maximum bytes can be kept in the payload field of a TCP segment ]

Max<sup>m</sup>

$$\text{TCP Segment Size} = \text{TCP Header size} + \text{MSS} [\text{Max}^m \text{ TCP Payload Size}]$$





# Topic : TCP

P<sub>1</sub>

TCP Client

TCP Server

P<sub>2</sub>

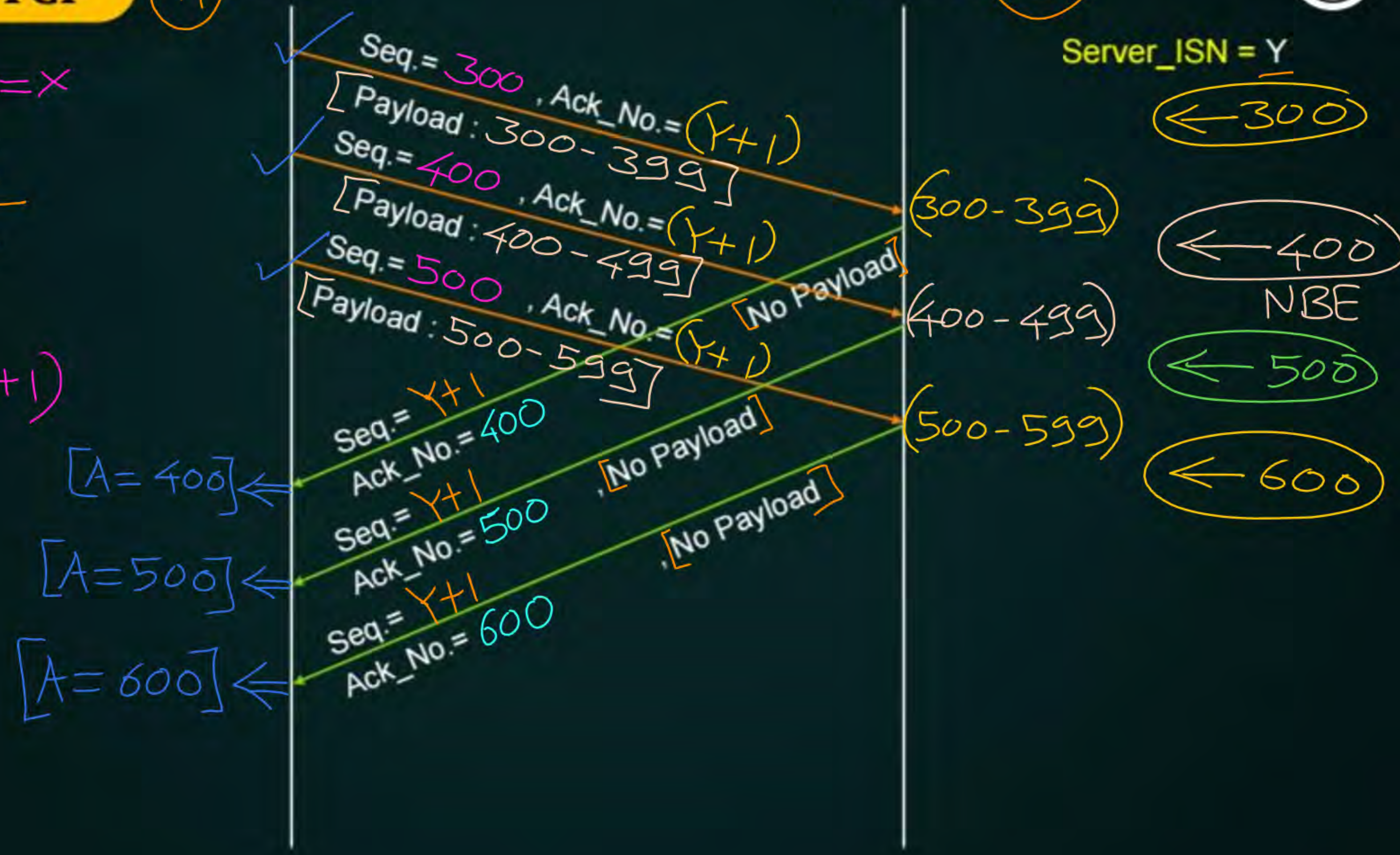


Client\_ISN = 299 = X

MSS = 100 bytes

First byte  
Seq. No.  
= 300 = (X+1)

Server\_ISN = Y







## Topic : TCP

Client\_ISN = 299

MSS = 100 bytes



TCP Client

TCP Server

Server\_ISN = Y

Seq.=300 [300-399], ACK = Y+1

Seq.=400 [400-499], ACK = Y+1

Seq.=500 [500-599], ACK = Y+1

Seq.=Y+1, ACK = 400

Seq.=Y+1, ACK = 500

Seq.=Y+1, ACK = 600



[MCQ] [GATE-IT-2004, 1 Mark]

IIT-D



#Q. In TCP, a unique sequence number is assigned to each :

☒ A byte

☐ B word

☐ C segment

☐ D message

TCP client

1<sup>st</sup> seg.  $\rightarrow$  Seq. No. = 299 (SYN)

2<sup>nd</sup> seg.  $\rightarrow$  Seq. No. = 300 (final ACK)

3<sup>rd</sup> seg.  $\rightarrow$  Seq. No. = ~~301~~ 400

4<sup>th</sup> seg.  $\rightarrow$  Seq. No. = ~~302~~ 500

Ans: A





## Topic : Sequence Number



=> Every byte in 'messages' is identified uniquely  
[with 32-bit sequence number]

Sequence Number in TCP Segment ← First Byte Sequence Number of Payload





## Topic : Sequence Number

TCP Socket → Socket\_Stream  
Stream Socket



⇒ TCP is byte-stream protocol (Stream-oriented protocol)

⇒ Unique sequence number is assigned to each byte





# Topic : TCP

Client\_ISN = 299

MSS = 100 bytes

TCP Client

TCP Server



Server\_ISN = Y

due to 300 combined ACK

(A = 400)  
(A = 500)

To 500

To 600

(ignore)  
(No action)

Seq. = 300 [300-399]

Seq. = 400 [400-499]

Seq. = 500 [500-599]

Seq. = 600 [600-699]

Seq. = 700 [700-799]

Ack\_No. = 400

Ack\_No. = 500

Duplicate ACK  
Ack\_No. = 500

Seq. = 500 [500-599]

Seq. = 600 [600-699]

Ack\_No. = 600

Ack\_No. = 800  
Combine ACK

300-399

400-499

(700-799)  
keep in Recv Buffer

(500-599)

(600-699)

← 300

← 400

← 500

← 500

← 600

← 800





## Topic : Acknowledgment Number



=> if ACK flag is on  
then TCP segment carry acknowledgment number  
else :  
Acknowledgment number is garbage value

(Combine ACK)

=> TCP uses cumulative acknowledgment  
[ ACK Number = NextByteExpected; ]

=> V. Imp

Combine ACK => The Ack acked more than one segment combinely



[MCQ] [GATE-2015, Set-3, 1 Mark]

IIT-K



#Q. Consider the following statements.

I. TCP connections are full duplex.

True  
Logical, Point-to-point, Full Duplex

II. TCP has no option for selective acknowledgment

(False)

III. TCP connections are message streams.

False (byte stream)

☒ A Only I is correct

☐ B Only I and II are correct

☐ C Only II and III are correct

☐ D All of I, II and III are correct

Ans: A





## Topic : Acknowledgment Number



=> In TCP, provision for 'selective acknowledgment'  
[ TCP keeps 'selective acknowledgment(s)' in optional header ]

TCP base header Ack → Compulsary  
Combine ACK

TCP optional header Ack → optional  
selective ACK





# Topic : TCP

Client\_ISN = 299

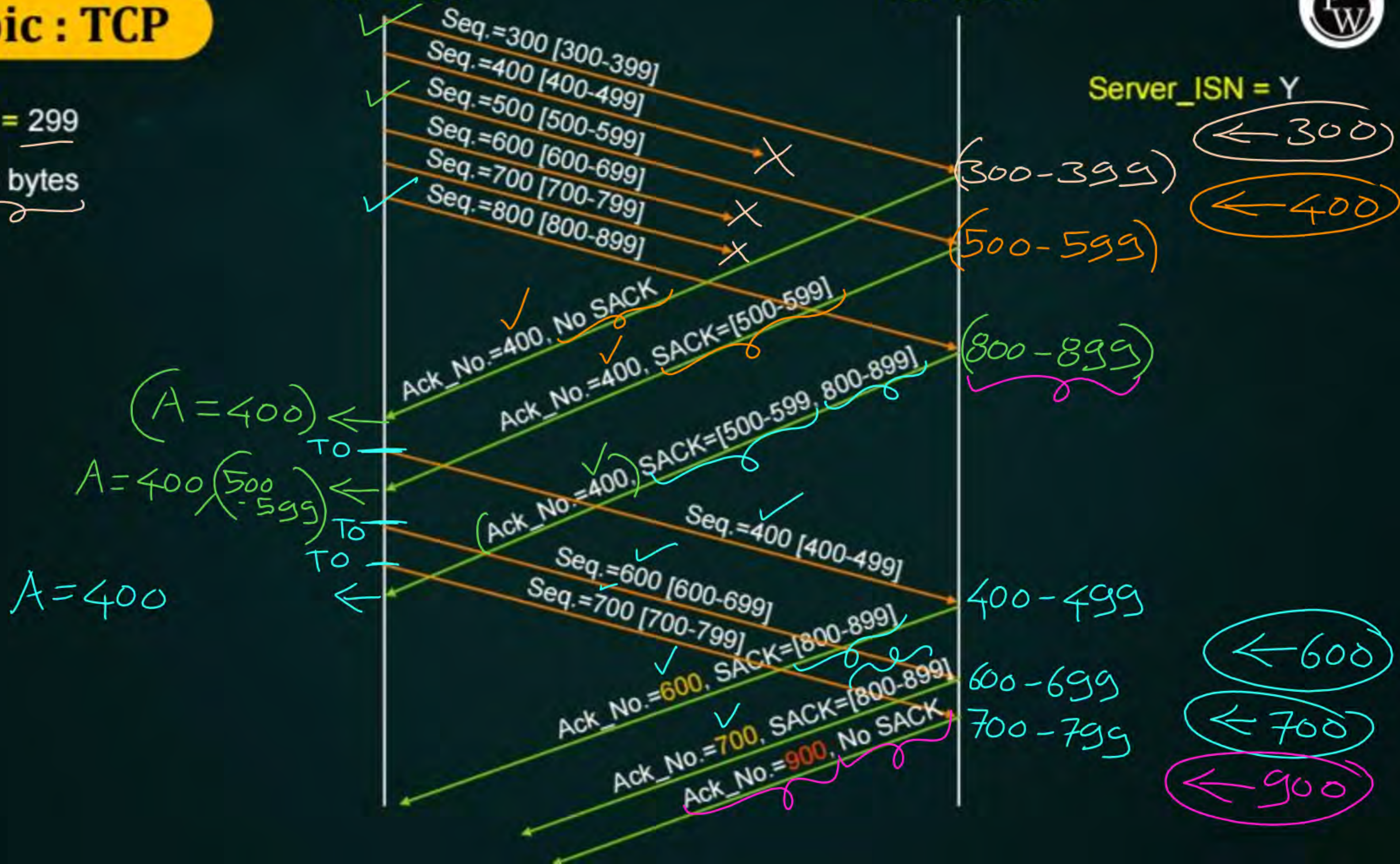
MSS = 100 bytes



TCP Client

TCP Server

Server\_ISN = Y





**[MCQ] [GATE-2025, Set-2, 1 Mark]**

IIT-R



#Q. Consider the following statements :

(i) Address Resolution Protocol (ARP) provides a mapping from an IP address to the corresponding hardware (link-layer) address.

(ii) A single TCP segment from a sender S to a receiver R cannot carry both data from S to R and acknowledgement for a segment from R to S.

Which ONE of the following is CORRECT?

- A** Both (i) and (ii) are TRUE
- B** (i) is TRUE and (ii) is FALSE
- C** (i) is FALSE and (ii) is TRUE
- D** Both (i) and (ii) are FALSE

(ii) → True or False





## Topic : Acknowledgment Number



=> TCP allows for piggy-backing  
[ sending data and acknowledgments in the same segment ]



### Example 11 :-

[NAT]



#Q. Consider a TCP connection where the sender sends two segments back to back. The sequence numbers (in decimal) of the first and second segments are 1300 and 2400 respectively. What is the size of data (in bytes) carried by the first segment in its payload field?

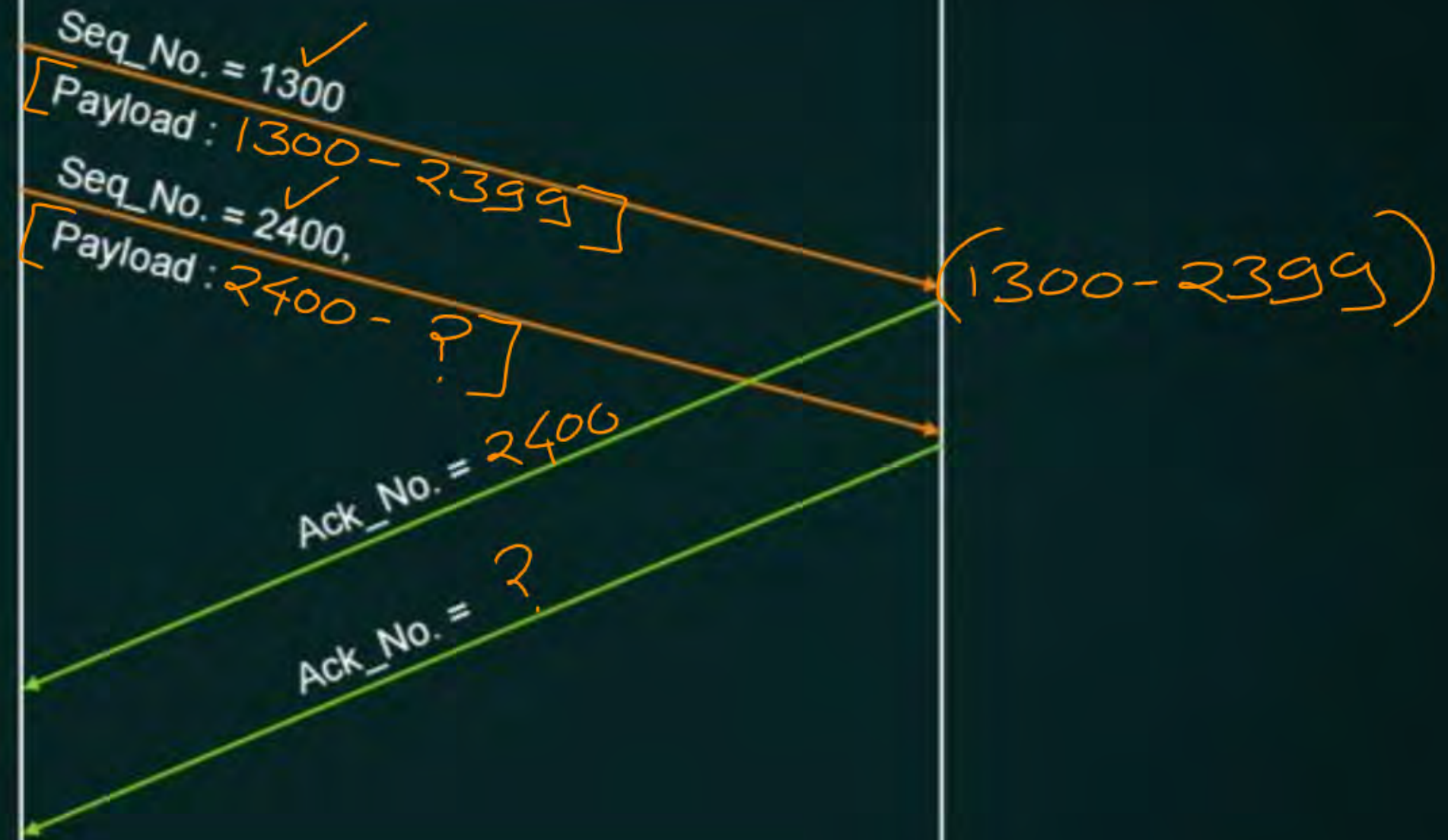
$$\text{Ans} = (2400 - 1300) = 1100$$

$$\boxed{\text{Ans} = 1100}$$



TCP Sender

TCP Receiver







## Summary



- 1 TCP Connection Establishment
- 2 TCP Data Transfer

TCP Congestion Control.



**THANK - YOU**